

Table 1 Counterfactual generation under PAIR (§1): per-feature activation on the original refused prompt vs. the jailbroken counterfactual (top-8 by SC plus the two lowest non-zero SC). a_f : activation; $\Delta\%$: activation change ($-$ = suppressed); γ : multiplicative change in suppression strength.

(a) Qwen3-8B							(b) Gemma-3-4B-IT							(c) Qwen3-1.7B						
Feature	SC	CAP	a_f orig	a_f cf	Δ %	γ $\times$	Feature	SC	CAP	a_f orig	a_f cf	Δ %	γ $\times$	Feature	SC	CAP	a_f orig	a_f cf	Δ %	γ $\times$
L31/F97078	0.538	0.726	44.1	19.9	-55%	+3.3 $\times$	L33/F305	0.825	-1.583	55.6	64.7	+16%	0.92 $\times$	L25/F137908	0.791	0.281	37.9	28.9	-24%	+1.7 $\times$
L35/F133412	0.499	0.850	51.7	20.9	-60%	+3.2 $\times$	L28/F763	0.717	0.353	48.3	32.2	-33%	+2.3 $\times$	L27/F53045	0.770	0.148	24.7	18.4	-26%	+2.0 $\times$
L29/F25883	0.430	0.184	20.4	22.4	+10%	+1.7 $\times$	L33/F453	0.636	-0.009	22.2	24.6	+11%	0.90 $\times$	L27/F124343	0.760	0.100	21.3	17.8	-16%	+1.6 $\times$
L35/F113160	0.387	0.552	40.1	23.1	-42%	+2.3 $\times$	L27/F113	0.599	0.179	30.8	23.8	-23%	+1.8 $\times$	L22/F19533	0.664	-0.000	17.3	15.2	-12%	+1.2 $\times$
L32/F108757	0.362	0.041	16.1	18.6	+16%	0.88 $\times$	L29/F563	0.583	0.277	38.7	27.9	-28%	+2.1 $\times$	L27/F37712	0.653	4.238	59.6	11.9	-80%	+4.0 $\times$
L35/F152241	0.355	0.249	28.6	20.7	-28%	+2.0 $\times$	L33/F695	0.566	0.014	18.2	19.6	+8%	+1.3 $\times$	L25/F77907	0.651	0.184	27.2	19.8	-27%	+1.3 $\times$
L35/F76102	0.266	0.120	22.4	18.3	-18%	+1.9 $\times$	L30/F588	0.555	0.036	26.4	24.2	-8%	+1.1 $\times$	L24/F150832	0.645	0.240	29.7	23.9	-20%	+2.2 $\times$
L35/F46295	0.258	0.327	30.9	20.9	-32%	+2.6 $\times$	L29/F369	0.549	0.102	32.7	27.9	-15%	+1.5 $\times$	L21/F1410	0.638	-0.460	19.3	22.3	+16%	0.85 $\times$
L35/F159803	0.248	0.423	23.7	25.7	+8%	0.90 $\times$	L33/F651	0.448	0.170	14.1	16.1	+14%	0.90 $\times$	L24/F132467	0.463	-0.006	14.9	12.8	-14%	+1.4 $\times$
L31/F68813	0.222	0.081	13.3	15.3	+15%	0.92 $\times$	L32/F581	0.422	0.002	10.7	12.6	+18%	0.88 $\times$	L24/F5220	0.437	-0.016	11.6	13.0	+12%	0.90 $\times$

Table 2 CAP components on WildJailbreak (§2): z_α, z_γ are z-normalised across each model’s validated features ($SC > 0$); $CAP = z_\alpha \cdot z_\gamma SC$; top-10 features by CAP. Validated-feature counts: Gemma-3-4B-IT 39, Qwen3-1.7B 50, Llama-3.2-1B 50.

(a) Gemma-3-4B-IT					(b) Qwen3-1.7B					(c) Llama-3.2-1B				
Feature	z_α	z_γ	SC	CAP	Feature	z_α	z_γ	SC	CAP	Feature	z_α	z_γ	SC	CAP
L33/F225	4.864	4.258	0.473	9.796	L27/F37712	4.790	1.355	0.653	4.238	L15/F33723	2.208	0.592	0.593	0.775
L33/F429	1.796	2.483	0.493	2.199	L23/F108420	2.592	1.720	0.481	2.144	L14/F121408	-0.485	-2.606	0.604	0.763
L33/F543	1.126	2.059	0.537	1.245	L27/F146528	1.762	0.990	0.577	1.007	L15/F32996	-0.803	-1.087	0.705	0.615
L28/F763	-0.719	-0.684	0.717	0.353	L27/F47793	1.587	0.596	0.481	0.455	L15/F86263	1.192	0.751	0.692	0.619
L29/F2139	0.554	1.412	0.495	0.387	L25/F137908	-0.830	-0.428	0.791	0.281	L15/F45856	-0.396	-1.806	0.733	0.524
L29/F563	-0.740	-0.643	0.583	0.277	L27/F162590	-0.388	-1.038	0.632	0.255	L14/F91437	-0.460	-1.407	0.602	0.390
L33/F2733	-0.641	-0.656	0.516	0.217	L24/F150832	-0.944	-0.394	0.645	0.240	L15/F60347	-0.396	-1.247	0.655	0.323
L27/F298	-0.596	-0.606	0.535	0.193	L22/F70992	-0.653	-0.627	0.560	0.229	L15/F115764	0.175	2.110	0.715	0.264
L30/F384	-0.592	-0.724	0.480	0.206	L25/F77907	-0.590	-0.478	0.651	0.184	L15/F113097	0.239	1.790	0.646	0.276
L27/F113	-0.633	-0.472	0.599	0.179	L22/F159911	-0.853	-0.350	0.613	0.183	L15/F48033	0.302	1.391	0.596	0.250

Table 3 SC-leakage probe AUROC (§4)

Model	Full CAP (150-d)	No-SC (100-d)	SC- perm.	SC-only (50-d)	Learn. SC wt.
Gemma-2-2B	0.788	0.788	0.788	0.500	0.000
Llama-3.2-1B	0.771	0.771	0.771	0.500	0.000
Qwen3-1.7B	0.940	0.940	0.940	0.500	0.000

Table 4 CSFD threshold sensitivity (§5)

Model	Spear- man ρ	Top-10 overlap	Candidate-set Jaccard (near default)
Gemma-3-4B-IT	1.00	1.00	0.79–1.00
Qwen3-1.7B	1.00	1.00	1.00 (0.54 @ 10 $\times$)
Llama-3.2-1B	1.00	1.00	1.00; 0.79–0.96 nearby

Table 5 Interventional validation of suppression edges (§6)

Intervention	Mean Δa_f	Refusal $\rightarrow$ compliance
Baseline (no patch)	0%	0.00
Suppressors amplified $\times 2$	-24%	0.09
Suppressors amplified $\times 4$	-35%	0.18
Suppressors amplified $\times 8$	-42%	0.29
Random same-layer $\times 8$ (control)	-6%	0.05

Table 6 LLM-judge agreement, % (§7); primary judge = gemini-3.1-flash-lite

Model	Judge pair	Jailbreak success	Harmful refused
Gemma-2-2B	primary vs gpt-5.4-mini	93.6	85.6
	primary vs Gemini-2.5-Pro	94.5	91.1
	gpt-5.4-mini vs Gemini-2.5-Pro	93.9	87.2
Qwen3-8B	primary vs gpt-5.4-mini	95.0	91.5
	primary vs Gemini-2.5-Pro	96.2	93.6
	gpt-5.4-mini vs Gemini-2.5-Pro	96.5	94.2